

Lecture 3

Principles of Quantum Mechanics

Art: *I'm not like you, Lenny. My brain just wasn't built for quantum mechanics*.

Lenny: *Nah, mine wasn't either. Just can't really visualize the stuff. But I'll tell you, I once knew a guy who thought just like an electron*.

Art: *What happened to him?*

Lenny: *Art, all I'm gonna tell you is that it sure wasn't pretty*.

Art: *Hmm, I guess that gene didn't fly*.

No, we were not built to sense quantum phenomena; not the same way we were built to sense classical things like force and temperature. But we are very adaptable creatures and we've been able to substitute abstract mathematics for the missing senses that might have allowed us to directly visualize quantum mechanics. And eventually we do develop new kinds of intuition.

This lecture introduces the principles of quantum mechanics. In order to describe those principles, we'll need some new mathematical tools. Let's get started.